

11(a)	any two points from: • (maximum) kinetic energy of electrons is independent of intensity • maximum kinetic energy of electrons depends on frequency • no time delay (between illumination and emission)	B2
11(b)(i)	(for $E_{\text{MAX}} = 0$,) $1/\lambda_0 = 1.93 \times 10^6 \text{ (m}^{-1}\text{)}$	C1
	$f_0 = 3.00 \times 10^8 \times 1.93 \times 10^6$ $= 5.8 \times 10^{14} \text{ Hz}$	A1
11(b)(ii)	$hc/\lambda = \Phi + E_{\text{MAX}}$	C1
	$hc = \text{gradient}$	C1
	gradient = e.g. $[(0.40 - 0.20) \times 1.60 \times 10^{-19}] / [(2.25 - 2.09) \times 10^6] \text{ (working needed)}$ $(= 2.0 \times 10^{-25})$	M1
	$h = (2.0 \times 10^{-25}) / (3.00 \times 10^8) = 6.7 \times 10^{-34} \text{ J s (both working and answer needed)}$	A1
11(c)	straight line with same gradient as the original	B1
	straight line with x-axis intercept greater than $1.93 \times 10^6 \text{ m}^{-1}$	B1